

ILOMANTSI MEKRIJARVI, Finland

WMO: 029390

Lat: 62.7697N

Lon: 30.9753E

Elev: 157

StdP: 99.46

Time Zone: 2.00 (EUE)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.3	(c) -24.7	(d) -31.4	(e) 0.2	(f) -28.3	(g) -27.3	(h) 0.3	(i) -24.6	(j) 9.3	(k) -1.0	(l) 8.4	(m) -1.5	(n) 1.1	(o) 30	(p) 0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	26.4	18.7	24.6	17.8	22.7	16.8	20.2	24.6	19.0	22.9	17.9	21.5	2.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.7	22.5	17.5	12.7	21.2	16.4	11.9	19.8	58.7	24.4	54.5	22.9	51.0	21.5	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.9	6.9	6.1	-30.5	28.5	4.1	2.7	-33.5	30.5	-35.9	32.1	-38.2	33.6	-41.2	35.6
			-30.6	21.6	4.1	1.7	-33.6	22.8	-35.9	23.7	-38.2	24.7	-41.2	25.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-10.0	-9.4	-5.0	1.7	9.1	13.2	16.9	14.9	9.8	3.2	-1.5	-5.3
	DBStd	10.41	7.86	7.14	5.34	3.82	4.54	3.90	3.55	3.41	3.15	3.97	4.55	6.40
	HDD10.0	3039	620	544	464	251	72	12	1	3	41	212	344	476
	HDD18.3	5564	878	778	722	500	288	159	71	116	256	469	594	734
	CDD10.0	559	0	0	0	1	43	109	214	155	36	1	0	0
	CDD18.3	41	0	0	0	0	1	6	26	9	0	0	0	0
	CDH23.3	303	0	0	0	0	16	48	183	56	0	0	0	0
	CDH26.7	45	0	0	0	0	0	6	32	7	0	0	0	0
Wind	WSAvg	2.8	2.6	2.8	3.0	2.9	3.0	2.7	2.4	2.4	2.7	3.1	3.3	3.1
Precipitation	PrecAvg	686	46	36	34	33	52	73	81	86	65	63	62	57
	PrecMax	916	82	62	56	70	98	113	167	141	116	121	101	88
	PrecMin	578	18	12	14	14	17	38	16	40	31	20	37	20
	PrecStd	88	17	16	11	15	26	22	40	30	24	26	18	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	2.8	7.9	15.2	25.4	27.2	29.6	27.5	19.7	12.4	8.0	5.5
		MCWB	1.8	1.6	3.7	8.2	14.7	17.9	20.7	19.7	15.6	11.4	7.4	5.1
	2%	DB	1.2	1.3	4.8	12.1	22.6	24.6	27.2	24.7	17.7	10.7	6.3	3.4
		MCWB	0.8	0.6	1.8	6.6	14.1	17.1	19.6	18.5	14.1	9.8	5.6	2.9
	5%	DB	0.5	0.5	2.9	9.8	19.6	22.4	25.3	22.7	15.9	9.6	5.1	2.2
		MCWB	0.2	0.1	0.9	5.0	12.6	15.7	18.9	17.3	13.4	8.8	4.7	1.8
	10%	DB	-0.1	-0.2	1.5	7.7	17.1	20.3	23.5	20.8	14.6	8.5	3.9	1.3
		MCWB	-0.3	-0.5	0.2	3.8	10.8	14.6	17.8	16.4	12.7	7.7	3.6	0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	1.7	4.1	9.7	17.1	20.2	22.1	20.9	16.6	11.7	7.4	5.1
		MCDB	2.3	2.2	7.6	13.1	22.3	24.6	27.1	26.0	18.6	12.2	7.8	5.5
	2%	WB	1.0	0.9	2.3	7.2	15.0	18.3	20.8	19.3	15.1	10.2	5.7	3.1
		MCDB	1.2	1.3	4.1	11.5	20.6	22.8	25.7	23.0	16.7	10.5	6.1	3.3
	5%	WB	0.3	0.2	1.2	5.5	13.4	16.7	19.7	18.2	14.0	9.0	4.8	1.9
		MCDB	0.5	0.6	2.3	8.9	18.4	20.7	23.9	21.6	15.5	9.4	5.2	2.1
	10%	WB	-0.4	-0.6	0.6	4.2	11.7	15.3	18.6	17.1	12.9	7.8	3.6	1.0
		MCDB	-0.1	-0.2	1.4	7.0	16.1	19.5	22.4	20.1	14.3	8.4	3.9	1.2
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.5	8.3	8.2	9.3	8.9	8.8	7.9	6.4	4.2	3.4	4.6
		MCDBR	3.1	4.4	8.3	11.5	14.0	11.8	11.3	11.0	8.7	5.0	3.7	3.4
		MCWBR	3.1	4.4	6.0	6.9	7.1	5.8	5.2	5.5	5.7	4.0	3.6	3.3
	5% WB	MCDBR	3.1	4.4	6.4	9.7	12.0	10.0	10.3	9.2	7.3	4.5	3.7	3.2
		MCWBR	3.1	4.3	4.9	6.3	6.7	5.5	5.2	5.1	5.2	4.0	3.6	3.1
Clear-Sky Solar Irradiance	taub	0.231	0.284	0.310	0.347	0.351	0.344	0.372	0.353	0.332	0.310	0.246	0.209	
	taud	2.005	2.106	2.158	2.277	2.443	2.488	2.425	2.460	2.509	2.399	2.096	1.963	
	Ebn at Noon	402	642	781	834	867	879	840	826	776	635	382	225	
	Edh at Noon	34	68	95	105	98	95	99	88	69	52	30	18	
All-Sky Solar Radiation	RadAvg	0.20	0.76	2.21	3.67	4.82	5.24	4.96	3.77	2.09	0.80	0.20	0.09	
	RadStd	0.03	0.17	0.28	0.38	0.53	0.34	0.48	0.41	0.26	0.11	0.04	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	-207	-203	N/A	N/A

Nomenclature: See separate page